Bias-Aware Job-Resume Matching: Auditing and Mitigating Demographic Bias in Automated Hiring Systems

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***Abstract*—Automated resume-screening systems are increasingly used to rank candidates before human review, but these systems may be sensitive to demographic cues that are unrelated to qualifications. In this project, we build a resume-job matching pipeline using Sentence-BERT and audit it with counterfactual resume variants. Each variant changes only one demographic signal, such as name, pronouns, or university, while keeping qualifications and experience fixed. We compute cosine similarity between resume and job embeddings, then measure how much the score changes across counterfactual versions. Our results show that name changes produced the largest average score shift, followed by university and pronoun changes. To reduce this sensitivity, we implement a blind-screening mitigation method that removes identity-related information before scoring. After blind screening, score differences are substantially reduced. We also include an open-source instruction-tuned language model as a qualitative comparison for resume-job match decisions. Overall, this project shows that counterfactual auditing is useful for detecting demographic sensitivity in resume-matching systems and that simple mitigation can reduce this effect.**

I. Introduction

Many job applications are now filtered by automated systems before a recruiter or hiring manager reviews them. These systems are often designed to improve efficiency by ranking resumes according to how well they match a job description. However, if the model is sensitive to information such as an applicant's name, pronouns, or university, then the system may unintentionally reproduce unfair patterns in hiring.

This project studies whether a resume-job matching model changes its score when only demographic-related signals are modified. The main goal is not to build a production hiring system, but to audit a simplified matching pipeline and understand where score differences may appear. We use Sentence-BERT as the main matching model because it can represent resumes and job descriptions as dense semantic embeddings. We then compute cosine similarity between each resume and job description.

The project has four main parts. First, we create resume-job matching scores using Sentence-BERT. Second, we perform a counterfactual fairness audit by comparing original resumes with variants where only one demographic signal is changed. Third, we implement blind screening as a mitigation method by removing identity-related information before scoring. Finally, we add an open-source instruction-tuned language model as a qualitative comparison, following feedback to examine a current LLM in addition to Sentence-BERT.

II. Problem Description and Motivation

Automated hiring systems are important to study because they operate in a high-stakes setting. A small difference in score can affect whether an applicant is selected for further review. Prior research has shown that hiring decisions can be affected by demographic cues. Bertrand and Mullainathan [1] found that resumes with white-sounding names received more callbacks than otherwise identical resumes with Black-sounding names. More recent work has also shown that language models can display bias in resume-screening tasks [4].

The motivation for this project is to bring these concerns into a practical machine learning pipeline. Instead of only discussing bias in theory, we build a matcher, generate counterfactual resume variants, and measure whether the model score changes when demographic cues are changed. If two resumes contain the same qualifications but receive different scores because of a changed name, pronoun, or university, that difference is a fairness concern.

III. Research Questions

This project addresses the following research questions:

1) Does a Sentence-BERT resume-job matching model assign different similarity scores to resumes that are identical except for one demographic signal?
2) Which signal causes the largest score difference: name, pronouns, or university?
3) Does blind screening reduce the score differences across counterfactual resume versions?
4) How can an open-source instruction-tuned language model provide a qualitative comparison to the Sentence-BERT matching pipeline?

These questions focus on both auditing and mitigation. The audit identifies whether demographic sensitivity exists, and the mitigation step tests whether removing identity-related information reduces that sensitivity.

IV. Related Work

This project builds on prior work in resume-job matching and algorithmic fairness. In person-job fit modeling, Qin et al. [2] proposed a neural approach for matching candidates to jobs. Sentence-BERT, introduced by Reimers and Gurevych [3], provides efficient sentence embeddings and is useful for semantic similarity tasks. We use Sentence-BERT as the backbone for the matching pipeline because it allows resumes and job descriptions to be compared using embedding similarity instead of exact keyword overlap.

The fairness motivation comes from both classic and recent work. Bertrand and Mullainathan [1] demonstrated that demographic signals in resumes can affect hiring outcomes. Wilson and Caliskan [4] studied gender, race, and intersectional bias in resume screening with language models. Our project is closer to an audit study: we construct controlled counterfactual variants and measure whether the model score changes when only one demographic signal is modified.

V. Dataset

The dataset contains resumes and job descriptions across five job domains:

- Software Engineering
- Finance
- Marketing
- Healthcare
- Education

The job file contains five job descriptions, one from each domain. The resume variants file contains 40 resume versions. Each base resume has an original version and counterfactual versions where one demographic signal is changed. The changed signals are name, pronouns, and university.

The important feature of the dataset is that the qualifications, skills, and experience are kept fixed across counterfactual versions. This means that when the score changes between the original resume and a modified version, the change can be attributed to the modified demographic signal.

VI. Methodology

A. Sentence-BERT Matching

The main matching model uses Sentence-BERT with the `all-MiniLM-L6-v2` model. For each resume-job pair, the resume text and job description are converted into embeddings. We then compute cosine similarity:

$$score(r, j) = \cos(E_r, E_j) \tag{1}$$

where E_r is the resume embedding and E_j is the job description embedding. A higher score means the resume is more semantically similar to the job description.

We compute scores for every resume variant against every job description. With 40 resume variants and 5 jobs, this produces 200 resume-job similarity scores.

B. Counterfactual Fairness Audit

For each resume and job pair, we compare the score of the original resume with the score of the counterfactual version. The score difference is defined as:

$$\Delta = score_{changed} - score_{original} \tag{2}$$

We also compute the absolute difference:

$$|\Delta| = |score_{changed} - score_{original}| \tag{3}$$

The absolute difference measures how much the score changed, regardless of whether it increased or decreased. We summarize these differences by changed signal: name, pronoun, and university.

C. Blind Screening Mitigation

To reduce demographic sensitivity, we implement blind screening. In this step, identity-related information is removed from the resume text before scoring. Specifically, we remove the applicant's name, pronouns, and university from each resume. The remaining text still contains qualification-related content such as degree, skills, and experience.

After creating blind-screened resume text, we run Sentence-BERT scoring again and repeat the same counterfactual comparison. If blind screening works, the score differences between original and changed versions should decrease.

D. LLM-Based Qualitative Comparison

To address the feedback about examining a current LLM, we also include an open-source instruction-tuned language model. We use FLAN-T5 as a qualitative comparison. Instead of using it as the main fairness metric, we ask the model to classify selected resume-job pairs as a strong match, partial match, or weak match.

The LLM component is included to show how a generative language model can support resume-job matching decisions. However, we keep Sentence-BERT as the main numerical model because its similarity scores are consistent and easier to compare across counterfactual resume variants.

VII. Experimental Results

A. SBERT Matching Results

The Sentence-BERT scoring pipeline successfully generated 200 resume-job similarity scores. Each score represents the semantic similarity between one resume variant and one job description. The scores were then used for the fairness audit.

B. Counterfactual Fairness Results

Table I shows the average absolute score difference for each changed demographic signal before mitigation.

Table I: Average absolute score difference before blind screening.

Changed Signal	Avg. Absolute Difference
Name	0.021594
Pronoun	0.006963
University	0.012347

The results show that changing the applicant's name caused the largest average score shift. University changes had the second-largest effect, while pronoun changes had the smallest average effect. Although the differences are not extremely large, they show that the matching score is not completely insensitive to demographic-related text changes.

C. Blind Screening Results

After blind screening, the average absolute score differences became much smaller. Table II shows the results after removing name, pronouns, and university from the resume text.

Table II: Average absolute score difference after blind screening.

Changed Signal	Avg. Absolute Difference
Name	0.000000
Pronoun	0.003617
University	0.000000

These results suggest that blind screening reduced the effect of demographic cues. Name and university differences were reduced to zero in this experiment. Pronoun-related differences were also reduced compared with the original audit.

D. LLM Comparison Results

The LLM comparison was used as a qualitative experiment. The model classified selected original resume-job pairs into match levels such as strong match, partial match, or weak match. This allowed us to examine how an instruction-tuned language model responds to the resume-job matching task.

The LLM component is useful because it produces a human-readable judgment. However, the output is qualitative and can vary depending on prompt wording. For this reason, we do not use the LLM output as the main fairness metric. Instead, it is included as an additional comparison to the SBERT-based scoring method.

VIII. Discussion

The results show that counterfactual auditing is a useful way to test whether a resume-matching model is sensitive to demographic signals. Because the resume variants are identical except for one changed signal, score differences provide evidence that the model is reacting to that signal or to the way it changes the surrounding text.

The name signal produced the largest average difference before mitigation. This is important because names are not qualification-related, but they may carry demographic associations. University changes also affected the score, which is more complicated. A university can be seen as part of a candidate's education history, but it can also act as a proxy for socioeconomic background or institutional prestige. Pronouns had the smallest effect, but the score difference was still measurable.

Blind screening reduced the counterfactual differences. This suggests that removing identity-related information can make the matching process less sensitive to demographic cues. However, blind screening is not a perfect solution. Removing university information may reduce bias, but it may also remove education-related context that some employers consider relevant. This shows the tradeoff between fairness and information preservation.

IX. Limitations

This project has several limitations. First, the dataset is small, so the results should be interpreted as an audit-style experiment rather than a general conclusion about real hiring systems. Second, the resume and job descriptions are simplified compared with real applicant tracking systems. Third, only one mitigation method, blind screening, was fully implemented. The original proposal also discussed data augmentation and adversarial debiasing, but those methods are left for future work.

The LLM component also has limitations. The instruction-tuned language model was used for qualitative match decisions, not for full numerical fairness analysis. Its outputs can be sensitive to prompt design, and it may not always give stable decisions across repeated runs or different wording.

X. Ethical Considerations

This project focuses on an ethical problem in automated hiring. If a model assigns different scores to equally qualified applicants because of demographic cues, then it can unfairly influence who receives an interview opportunity. Even small score differences can matter when systems rank many applicants.

At the same time, this project does not claim to solve hiring bias. Names, pronouns, and universities are only proxies for demographic and social signals. Real identity and lived experience are more complex than these variables can represent. Blind screening can reduce some demographic sensitivity, but it does not remove deeper structural inequalities in hiring.

The project should therefore be understood as a fairness audit and mitigation study. The goal is to identify where score differences occur and test whether a simple mitigation step can reduce them.

XI. Future Work

Future work can extend this project in several ways. First, the dataset can be expanded with more resumes, job descriptions, and industries. Second, data augmentation can be implemented by training or evaluating the model with more counterfactual resume pairs. Third, adversarial debiasing can be tested to reduce demographic information in the learned representation. Fourth, stronger LLMs can be evaluated more systematically for both match quality and bias. Finally, future work can compare fairness improvements with potential changes in matching accuracy.

XII. Conclusion

This project built a bias-aware resume-job matching pipeline using Sentence-BERT, counterfactual fairness analysis, blind-screening mitigation, and an open-source LLM comparison. The results show that changing demographic signals can affect matching scores, especially for names and universities. Blind screening reduced these differences, suggesting that removing identity-related information can help reduce demographic sensitivity in this dataset. The LLM comparison provides an additional qualitative view of resume-job fit, while Sentence-BERT remains the main model for consistent numerical fairness analysis.

References

[1] Marianne Bertrand and Sendhil Mullainathan. Are emily and greg more employable than lakisha and jamal? a field experiment on labor market discrimination. *American Economic Review*, 94(4):991–1013, 2004.

[2] Chuan Qin, Hengshu Zhu, Tong Xu, Chen Zhu, Liang Jiang, Enhong Chen, and Hui Xiong. Enhancing person-job fit for talent recruitment: An ability-aware neural network approach. In *Proceedings of the 41st International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 25–34, 2018.

[3] Nils Reimers and Iryna Gurevych. Sentence-bert: Sentence embeddings using siamese bert-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing*, pages 3982–3992, 2019.

[4] Kyra Wilson and Aylin Caliskan. Gender, race, and intersectional bias in resume screening via language model retrieval. In *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*, volume 7, pages 1578–1590, 2024.